

DHRUV SINGH

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SUMMARY OF QUALIFICATIONS

- Full-stack developer (React, Node.js, OpenAI API) who launched Brainstormzz to **50+ beta users** in 2 weeks with a **98% successful generation rate**.
- Built **8 D2L course shells** in HTML/CSS and provisioned **500+ student accounts** as a Co-op Program Assistant at TMU.
- Built Python test automation across **100+ scenarios**, resolving **15+ defects** through regression testing.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C#, SQL, VHDL, HTML, CSS

Web & Tools: React, Node.js, REST API Integration, OpenAI API, Git, GitHub, Jira, Figma, Linux (command line), Excel

Testing & Hardware: Test Automation, Regression Testing, OpenCV, FPGA (Quartus, ModelSim), KiCad, ngspice

EDUCATION

Toronto Metropolitan University

Toronto, Canada

Bachelor of Engineering, Computer Engineering – Software Engineering Option (Co-op)

Sept 2023 – Apr 2028

WORK EXPERIENCE

Co-op Program Assistant

Toronto, Canada

Career, Co-op & Student Success Centre, TMU

Sept 2025 – Aug 2026

- Built **8 custom D2L (Brightspace) course shells** in HTML/CSS as the front-end for co-op programs, deployed to **500+ students**.
- Migrated and provisioned **500+ student accounts** with data mapping and access configuration, with zero reported access issues.
- Managed and validated employment records for **200+ co-op students** in Excel (VLOOKUP, advanced formulas), maintaining **95%+ data integrity**.

UI Technician

Toronto, Canada

Game Makers Union, TMU

Sept 2024 – Apr 2025

- Delivered **8 UI components** across 3 Agile sprints with a 10+ member cross-functional team, using Git for version control.
- Analyzed feedback from **25+ playtest sessions**, increasing task completion rate by **35%** and reducing average interaction time by **20%**.
- Profiled and resolved performance bottlenecks through systematic debugging, reducing frame drops by **40%**.

PROJECT EXPERIENCE

Brainstormzz – AI-Powered Idea Structuring Platform | React, Node.js, OpenAI API | GitHub

Dec 2025

- Built a full-stack AI platform turning unstructured input into visual maps, onboarding **50+ beta users** in 2 weeks.
- Ran user research with **12** early-stage founders, translating pain points into prioritized product requirements.
- Architected error handling and retry logic achieving a **98% successful generation rate**, reducing average time-to-insight by **60%**.

Rover Arm Controller & CV Test Automation | Python, OpenCV | GitHub

Nov 2025 – Dec 2025

- Designed a socket-based Python control system sending real-time motor commands to a **6-motor** robotic arm.
- Built a computer vision test automation framework in Python validating ARUCO tag detection across **100+ test scenarios**.
- Identified and resolved **15+ defects** through systematic regression testing and root-cause analysis.

8-Bit General-Purpose Processor | VHDL, FSM, FPGA | GitHub

Nov 2025 – Dec 2025

- Designed an **8-bit** ALU and FSM-based control unit in VHDL with precise timing constraints.
- Integrated register file, latches, and tri-state bus architecture into the **8-bit** processor datapath.
- Validated via simulation and FPGA deployment with a **100%** simulated-to-hardware match across all instruction types.

Carpool Matching Optimization System | Python | GitHub

Oct 2024

- Designed a route optimization algorithm matching **100+ drivers and riders** across 15 zones, cutting commute distance **22%**.
- Implemented multi-constraint matching logic with **100%** rule compliance across **50+** test scenarios.
- Built an evaluation framework scoring matches on **2** metrics, schedule alignment and CO₂ reduction, to guide matching decisions.

LEADERSHIP EXPERIENCE

- **TMU Engineering Student Ambassador** – Computer Engineering Representative, Ontario University Fair (Sept 2026)
- **Sponsorship Committee**, Thrill Club, TMU

AWARDS & ACCOMPLISHMENTS

- **3rd Place**, TMU First-Year Engineering Student Design Competition – 1 of 8 finalists, “Sustainable Eco Powered House” (Nov 2023)
- **The Complete Full-Stack Web Development Bootcamp** – Udemy, Dr. Angela Yu (61.5 hours, Feb 2025)